



Synthesis of xanthene derivatives, molecular docking and molecular dynamics simulations, drug similarity analysis, ADMET prediction, and assessment of carbonic anhydrase inhibitory activity

Gönül Bükülmez¹ · Şüheda Özkul² · Ekrem Tunca² · Alpaslan Bayrakdar³ · Rahmi Kasımoğulları¹

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Abstract

A series of xanthene derivatives (**3a–n**) was synthesized using microwave-assisted catalysis with trifluoroacetic acid (TFA) in an aqueous medium, achieving high yields within 10 min. The structural features of these compounds were comprehensively characterized by FT-IR, ¹H-NMR, ¹³C-NMR, and mass spectrometry (MS). The inhibitory effects of the synthesized xanthene derivatives on human carbonic anhydrase isoenzymes (hCA I and hCA II) were assessed in vitro. Notably, compounds **3c** and **3n** demonstrated significant inhibition of these isoenzymes, with compound **3c** exhibiting a K_i value of 3.619 μ M for hCA I and compound **3n** showing a K_i value of 0.178 μ M for hCA II. In silico analyses further supported the strong inhibitory activity of compounds **3c** and **3n** against hCA I and hCA II. Drug-likeness and ADMET properties were evaluated according to Lipinski's rules and computational prediction tools. Molecular docking studies indicated that both compounds formed strong interactions with their target receptors (2CAB and 1CA2) via active-site residues, including His94, His96, and His119. The binding affinities of **3c** and **3n** exceeded that of the reference inhibitor, acetazolamide (AAZ). Additionally, 100 ns molecular dynamics simulations confirmed the structural stability of the resulting complexes, as assessed by RMSD and RMSF analyses.

Keywords Carbonic anhydrase · Inhibition · Microwave · Molecular docking · Molecular dynamics · Xanthene derivatives

Introduction

1,8-dioxo-octahydroxanthenes represent a class of biologically significant heterocyclic compounds characterized by a xanthene skeleton. The distinguishing structural feature of xanthenes, compared to other oxygen-containing tricyclic compounds, is

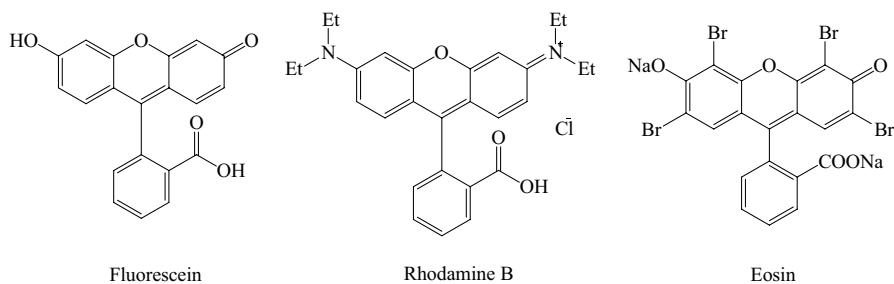


Fig. 1 Xanthene-based dyes

the presence of a dibenzo[*b,e*]pyran nucleus [1]. Xanthene and its derivatives are recognized as promising biologically active compounds with considerable pharmaceutical potential [2–4]. These molecules exhibit a range of biological activities, including anticancer [5], anti-inflammatory [6], anti-hyperlipidemic [7], antibacterial [8], antioxidant [9], and nematocidal effects [10]. Xanthene derivatives also constitute a major class of organic dyes, such as fluorescein, rhodamine B, and eosin (Fig. 1). These dyes are extensively utilized in biological imaging, photodynamic therapy, and various industrial applications due to their high molar absorption coefficients, favorable water solubility, low cytotoxicity, and elevated fluorescence quantum yields [11–14]. In addition to synthetic applications, the xanthene skeleton is found in several natural products, including Gaudichaudione A, Gaudichaudione B, Allanxanthone C, and Funiculosone (Fig. 2) [15–19].

Several methods have been developed for the synthesis of xanthene derivatives, and these approaches are generally evaluated based on environmental impact, efficiency, selectivity, and reaction time. Traditionally, acidic catalysts and organic solvents were used for synthesis. For example, [Hbim]BF₄ [20], *n*-TSA [21], SiO₂/H₁₄[NaP₅W₃₀O₁₁₀] [22], Amberlite™ IRA67 [23], BF₃·OEt₂ [24], citric acid-DMU [25], PEG-SO₃H [26], SmCl₃ [27], Selectfluor™ [28], [Msim]Cl [29], and Fe₃O₄@SiO₂-imide-PMA_n [30] have been reported as effective agents. However, these traditional methods often suffer from drawbacks such as long reaction times, low selectivity, and negative environmental impacts. In response, solvent-free, microwave-assisted, and heterogeneously catalyzed approaches have gained significant momentum, as they align with green chemistry principles and are increasingly used for the synthesis of heterocyclic compounds, particularly xanthene derivatives [31–34].

The carbonic anhydrase enzyme (CA, EC 4.2.1.1) is expressed in a wide range of prokaryotic and eukaryotic organisms and catalyzes the reversible hydration of carbon dioxide (CO₂) to bicarbonate (HCO₃[−]) and protons (H⁺) [35–37]. CA is encoded by eight distinct gene families (α, β, γ, δ, ζ, η, θ, ι) [38]; in mammals, it belongs to the α-gene family and contains a Zn²⁺ ion in its active site [39]. In humans, 15 different CA isoenzymes have been identified, each participating in diverse physiological, metabolic, and pathological processes [40], including respiration [41], acid–base balance [42], electrolyte secretion [43], metabolic reactions (gluconeogenesis, lipogenesis, ureagenesis) [44], bone resorption [45], and

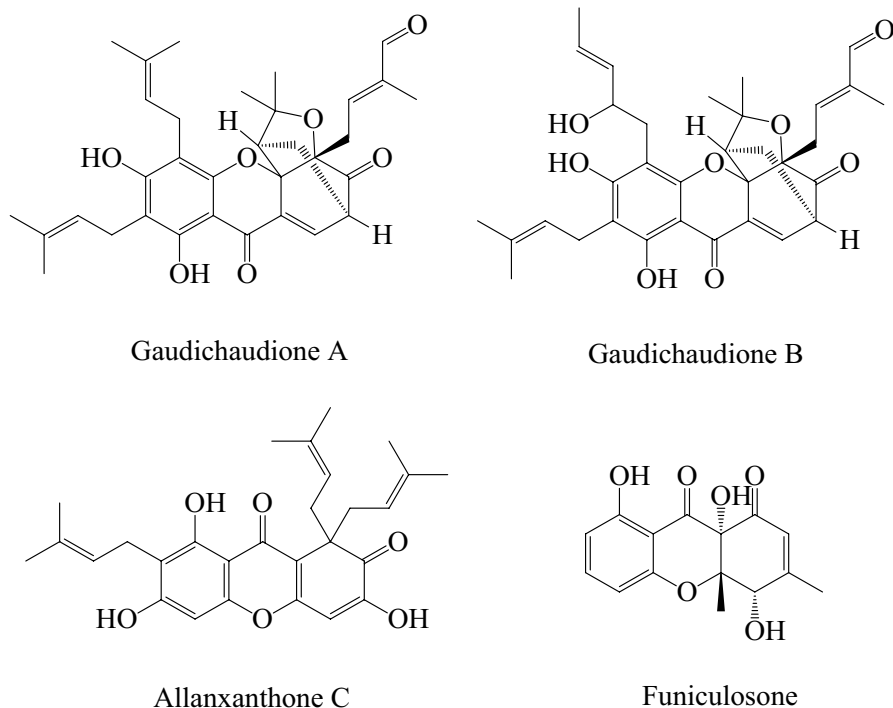


Fig. 2 Natural xanthenes

tumorigenicity [46]. The involvement of CA isoenzymes in numerous biological events underlies their association with various disorders. For example, hCA I is linked to glaucoma and retinal or cerebral edema [47]; hCA II to glaucoma, altitude sickness, and epilepsy [48]; hCA IV to glaucoma [49]; hCA VII to epilepsy [50]; and hCA IX and XII to cancer [51]. Additional studies suggest that hCA VA and VB may be associated with obesity [52], hCA VII with Alzheimer's disease [53], and hCA III with oxidative stress [54]. Recent findings indicate that hCA I may also be implicated in tumor formation, differentiation [55], and ankylosing spondylitis [56], while hCA II has been associated with certain brain tumors [57]. Although the precise mechanisms remain unclear, inhibition of these isoenzymes is a principal strategy for treating hCA-related disorders or reducing their impact [58].

Currently, sulfonamide derivatives, including AAZ, methazolamide, ethoxzolamide, dorzolamide, brinzolamide, and zonisamide, as well as the sulfamate derivative topiramate, are used clinically as hCA inhibitors [59]. However, these agents lack selectivity for specific hCA isoforms and can affect multiple hCAs, resulting in numerous side effects [60]. To address this limitation, researchers have developed hCA inhibitors from diverse compound classes, including coumarins [61], sulfo-coumarins [62], phenols [63], polyamines [64], benzoxaboroles [65], and selenols [66]. Despite these advances, broadening the chemical diversity of hCA inhibitors remains an important goal.

Computational (in silico) methods are essential in contemporary drug discovery, facilitating the evaluation of pharmacokinetic properties, target-binding affinities, and structural stability of candidate molecules. In this study, xanthene derivatives were synthesized in high yields using a TFA-catalyzed, microwave-assisted protocol in an aqueous medium. The inhibitory effects of the synthesized compounds on hCA I and hCA II were evaluated in vitro. Compounds **3c** and **3n**, which exhibited the highest inhibitory activity against hCA I and hCA II, were selected for further computational analysis. Initially, drug-likeness and ADMET profiles were assessed for these compounds. Subsequently, molecular docking studies were conducted to elucidate their binding modes within the active sites of the target enzymes, enabling identification of key interactions with critical amino acid residues. In our previous research we provided a comprehensive molecular docking analysis of the standard inhibitor AAZ in complex with both hCA I and hCA II, establishing a robust reference for comparison with the newly investigated ligands, **3c** and **3n** [67]. AAZ-related results obtained with the same validated methodology as a reference, without reanalysis were utilized in the present study. To further substantiate the findings, 100-ns molecular dynamics simulations were performed to assess the stability of the ligand–protein complexes. Overall, this integrated in silico approach highlights the value of computational methods in early-stage drug development and provides preliminary insights into the therapeutic potential of the synthesized molecules.

Experimental section

Materials and methods

All chemicals utilized in the experiments were obtained from commercial suppliers, including Merck, Sigma-Aldrich, Acros Organics, and Alfa-Aesar. These chemicals were of analytical reagent grade and were used without further purification. Microwave-assisted reactions were carried out in closed reaction vessels using a CEM Discover SP mono-mode reactor (CEM Corp., Matthews, NC, USA).

The progress of all synthesized intermediates was monitored by thin-layer chromatography (TLC) on pre-coated silica gel 60 F254 aluminum plates (Merck, Darmstadt, Germany). TLC plates were examined under ultraviolet light at 254 and 366 nm with a TLC imager (Camag, Upland, CA, USA). Melting points were determined using a Stuart Automatic Melting Point Apparatus SMP50 (Staffordshire, UK) by recording both the initial and final melting temperatures. Fourier-transform infrared (FT-IR) spectra were obtained with a Bruker Vertex 70 FT-IR spectrometer using the potassium bromide (KBr) pellet method.

¹H-NMR and ¹³C-NMR spectra were acquired using a Bruker Avance III 400 MHz NMR spectrometer (Bruker Optics Inc., MA, USA). MS data were collected with an Agilent 6530 Accurate-Mass quadrupole time-of-flight (Q-TOF) instrument (Agilent, Santa Clara, CA). Enzyme activities were measured using a Shimadzu UV-1700 PharmaSpec spectrophotometer (Shimadzu, Tokyo, Japan).

General procedure for xanthene derivatives (3a–n)

Substituted benzaldehyde derivatives (**1a–n**) (1 mmol) and cyclic-1,3-diketone derivatives (**2a–c**) (2 mmol) were each dissolved in approximately 15 mL of pure water in separate containers to obtain homogeneous solutions. The solutions were then combined, and 0.5 mmol TFA was added as a catalyst. The resulting mixture was subjected to microwave irradiation for 10 min (reaction conditions: 17 atm, 200 W, and 100 °C). Upon completion of the reaction, the mixture was transferred into ice water without prior cooling. The resulting precipitate was collected by filtration and recrystallised from a suitable solvent to yield the pure product.

9-(*p*-tolyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (**3a**)

Compound **3a** was synthesized by reacting 4-methylbenzaldehyde (**1a**) (0.118 mL, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from ethanol to yield white crystals. Yield 339.97 mg, 93%; mp: 214–216 °C; IR (KBr, ν , cm^{-1}): 3046 (aromatic-CH), 2960 and 2882 (aliphatic-CH), 1659 (C=O), 1621, 1512, 1356, 1201 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 7.17 (d, 2H, $J=7.9$ Hz, Ar–H), 7.01 (d, 2H, $J=7.8$ Hz, Ar–H), 4.70 (s, 1H, pyran-CH), 2.45 (s, 3H, Ar-CH₃), 2.12–2.24 (m, 8H, $4\times\text{CH}_2$), 1.08 (s, 6H, $2\times\text{CH}_3$), 0.98 (s, 6H, $2\times\text{CH}_3$), $^{13}\text{C-NMR}$ (100 MHz, CDCl_3) δ_{C} (ppm): 196.73 (C=O), 162.39 (C_{4a}, C_{10a}), 141.43, 135.97, 129.017, 128.46, 115.94, 50.98 (C₂, C₇), 41.06 (C₄, C₅), 32.44 (C₉), 31.66 (C₃, C₆), 29.53 and 27.59 (CH₃), 21.32 (Ar-CH₃); MS (ESI): m/z $[\text{M} + \text{H}]^+$ calcd. for C₂₄H₂₈O₃: 364.20; Found: 365.21.

9-(4-methoxyphenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (**3b**)

Compound **3b** was synthesized by reacting 4-methoxybenzaldehyde (**1b**) (0.121 mL, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from ethanol to yield white crystals. Yield 350.042 mg, 92%; mp: 246–248 °C; IR (KBr, ν , cm^{-1}): 3059 (aromatic-CH), 2963 and 2869 (aliphatic-CH), 1671 (C=O), 1627, 1508, 1359, 1265 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 7.19 (d, 2H, $J=8.5$ Hz, Ar–H), 6.74 (d, 2H, $J=8.5$ Hz, Ar–H), 4.68 (s, 1H, pyran-CH), 3.71 (s, 3H, OCH₃), 2.12–2.44 (m, 8H, $4\times\text{CH}_2$), 1.08 (s, 6H, $2\times\text{CH}_3$), 0.97 (s, 6H, $2\times\text{CH}_3$), $^{13}\text{C-NMR}$ (100 MHz, CDCl_3) δ_{C} (ppm): 196.80 (C=O), 162.34 (C_{4a}, C_{10a}), 158.12 (phenyl-C–O), 136.70, 129.52, 115.96, 113.65, 55.32 (OCH₃), 50.97 (C₂, C₇), 41.04 (C₄, C₅), 32.43 (C₉), 31.17 (C₃, C₆), 29.53 and 27.55 (CH₃); MS (ESI): m/z $[\text{M} + \text{Na}]^+$ calcd. for C₂₄H₂₈O₄: 380.48; Found: 403.18.

9-(3,4-dimethoxyphenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (3c)

Compound **3c** was synthesized by reacting 3,4-dimethoxybenzaldehyde (**1c**) (0.166 g, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from ethanol to yield white crystals. Yield 369.189 mg, 90%; mp: 182–184 °C; IR (KBr, ν , cm^{-1}): 3083 (aromatic-CH), 2963 and 2872 (aliphatic-CH), 1668 (C=O), 1622, 1463, 1359, 1263 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 6.68–6.88 (m, 3H, Ar–H), 4.68 (s, 1H, pyran-CH), 3.83 (s, 3H, OCH_3), 3.77 (s, 3H, OCH_3), 2.18–2.44 (m, 8H, $4 \times \text{CH}_2$), 1.08 (s, 6H, $2 \times \text{CH}_3$), 0.98 (s, 6H, $2 \times \text{CH}_3$), $^{13}\text{C-NMR}$ (100 MHz, CDCl_3) δ_{C} (ppm): 196.82 (C=O), 162.41 ($\text{C}_{4\text{a}}$, $\text{C}_{10\text{a}}$), 148.58, 147.60, 137.17, 120.29, 115.92, 112.39, 110.96, 56.06 (OCH_3), 55.94 (OCH_3), 50.95 (C_2 , C_7), 41.06 (C_4 , C_5), 32.42 (C_9), 31.43 (C_3 , C_6), 29.58 and 27.45 (CH_3); MS (ESI): m/z [$\text{M} + \text{Na}$] $^+$ calcd. for $\text{C}_{25}\text{H}_{30}\text{O}_5$: 410.21; Found: 433.20.

9-(3-ethoxy-4-hydroxyphenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (3d)

Compound **3d** was synthesized by reacting 3-ethoxy-4-hydroxybenzaldehyde (**1d**) (0.166 g, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from a mixture of ethanol and water to yield yellow crystals. Yield 377.39 mg, 92%; mp: 194–196 °C; IR (KBr, ν , cm^{-1}): 3435, 3030 (aromatic-CH), 2954 and 2883 (aliphatic-CH), 1674 (C=O), 1513, 1440, 1354 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 6.94 (d, 1H, $J=2.0$ Hz, Ar–H), 6.69 (d, 1H, $J=7.5$ Hz, Ar–H), 6.54 (dd, 1H, $J=7.5$, 2.0 Hz, Ar–H), 5.70 (s, 1H, Ar–OH), 4.61 (s, 1H, pyran-CH), 4.08 (q, $J=7.0$ Hz, 2H, OCH_2CH_3), 2.16–2.42 (m, 8H, $4 \times \text{CH}_2$), 1.35 (t, $J=7.0$ Hz, 3H, OCH_2CH_3), 1.06 (s, 6H, $2 \times \text{CH}_3$), 0.96 (s, 6H, $2 \times \text{CH}_3$), $^{13}\text{C-NMR}$ (100 MHz, CDCl_3) δ_{C} (ppm): 196.88 (C=O), 162.35 ($\text{C}_{4\text{a}}$, $\text{C}_{10\text{a}}$), 145.38, 144.31, 136.51, 120.14, 115.97, 114.11, 113.30, 64.48 (OCH_2CH_3), 50.96 (C_2 , C_7), 41.02 (C_4 , C_5), 32.41 (C_9), 31.49 (C_3 , C_6), 29.53 and 27.48 (CH_3), 15.08 (OCH_2CH_3); MS (ESI): m/z [$\text{M} + \text{Na}$] $^+$ calcd. for $\text{C}_{25}\text{H}_{30}\text{O}_5$: 410.21; Found: 433.202.

9-(4-hydroxy-3-methoxyphenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (3e)

Compound **3e** was synthesized by reacting 4-hydroxy-3-methoxybenzaldehyde (**1e**) (0.155 g, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from a mixture of ethanol and water to yield white crystals. Yield 360.13 mg, 91%; mp: 224–226 °C; IR (KBr, ν , cm^{-1}): 3416 (Ar–OH), 3036 (aromatic-CH), 2963 and 2858 (aliphatic-CH), 1660 (C=O), 1517, 1368, 1277 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 6.97 (s, 1H, Ar–H), 6.70 (d, 1H, $J=8.1$ Hz, Ar–H), 6.56 (d, 1H, $J=7.9$ Hz,

Ar-H), 5.67 (s, 1H, Ar-OH), 4.64 (s, 1H, pyran-CH), 3.84 (s, 3H, OCH₃), 2.18–2.44 (m, 8H, 4×CH₂), 1.07 (s, 6H, 2×CH₃), 0.98 (s, 6H, 2×CH₃), ¹³C-NMR (100 MHz, CDCl₃) δ_C (ppm): 196.97 (C=O), 162.39 (C_{4a}, C_{10a}), 146.13, 144.22, 136.62, 120.26, 115.98, 114.99, 112.43, 56.06 (OCH₃), 50.95 (C₂, C₇), 41.04 (C₄, C₅), 32.43 (C₉), 31.50 (C₃, C₆), 29.54 and 27.47 (CH₃); MS (ESI): *m/z* [M+H]⁺ calcd. for C₂₄H₂₈O₅: 396.19; Found: 397.203.

9-(4-isopropylphenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (3f)

Compound **3f** was synthesized by reacting 4-isopropylbenzaldehyde (**1f**) (0.1523 mL, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from a mixture of ethanol and water to yield yellow crystals. Yield 349.093 mg, 89%; mp: 170–172 °C; IR (KBr, ν, cm⁻¹): 3034 (aromatic-CH), 2963 and 2870 (aliphatic-CH), 1664 (C=O), 1625, 1459, 1366, 1193 (C=C, C-O); ¹³C-NMR (100 MHz, CDCl₃) δ_C (ppm): 196.83 (C=O), 162.45 (C_{4a}, C_{10a}), 146.70, 141.57, 128.32, 126.35, 116.00, 50.99 (C₂, C₇), 41.07 (C₄, C₅), 33.82 (CH(CH₃)₂), 32.45 (C₉), 31.50 (C₃, C₆), 29.46 and 27.69 (CH₃), 24.15 (CH(CH₃)₂); MS (ESI): *m/z* [M+H]⁺ calcd. for C₂₆H₃₂O₃: 392.24; Found: 393.24.

9-(4-(methylthio)phenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (3g)

Compound **3g** was synthesized by reacting 4-(methylthio)benzaldehyde (**1g**) (0.133 mL, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from a mixture of ethanol and water to yield white crystals. Yield 360.52 mg, 91%; mp: 209–211 °C; IR (KBr, ν, cm⁻¹): 3030 (aromatic-CH), 2963 and 2870 (aliphatic-CH), 1662 (C=O), 1622, 1471, 1361, 1195 (C=C, C-O); ¹H-NMR (400 MHz, CDCl₃) δ_H (ppm): 7.19 (d, 2H, *J*=8.3 Hz, Ar-H), 7.09 (d, 2H, *J*=8.3 Hz, Ar-H), 4.67 (s, 1H, pyran-CH), 2.39 (s, 3H, S-CH₃), 2.12–2.44 (m, 8H, 4×CH₂), 1.08 (s, 6H, 2×CH₃), 0.97 (s, 6H, 2×CH₃), ¹³C-NMR (100 MHz, CDCl₃) δ_C (ppm): 196.76 (C=O), 162.57 (C_{4a}, C_{10a}), 141.47, 136.08, 129.09, 126.67, 115.65, 50.92 (C₂, C₇), 41.04 (C₄, C₅), 32.44 (C₉), 31.60 (C₃, C₆), 29.52 and 27.53 (CH₃), 16.09 (S-CH₃); MS (ESI): *m/z* [M+Na]⁺ calcd. for C₂₄H₂₈O₃S: 396.18; Found: 419.16.

9-(4-bromophenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (3h)

Compound **3h** was synthesized by reacting 4-bromobenzaldehyde (**1h**) (0.185 g, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from a mixture of ethanol and water to yield white crystals. Yield 398.133 mg, 93%; mp: 240–242 °C; IR (KBr, ν, cm⁻¹): 3038 (aromatic-CH), 2959 and 2875 (aliphatic-CH), 1657 (C=O),

1620, 1468, 1359, 1195 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 7.39 (d, 2H, $J=6.7$ Hz, Ar–H), 7.10 (d, 2H, $J=6.0$ Hz, Ar–H), 4.69 (s, 1H, pyran-CH), 2.04–2.53 (m, 8H, $4\times\text{CH}_2$), 1.01 (s, 6H, $2\times\text{CH}_3$), 0.87 (s, 6H, $2\times\text{CH}_3$), $^{13}\text{C-NMR}$ (100 MHz, CDCl_3) δ_{C} (ppm): 196.81 (C=O), 163.75 ($\text{C}_{4\text{a}}$, $\text{C}_{10\text{a}}$), 147.46, 131.48, 131.44, 131.05, 114.57, 50.60 (C_2 , C_7), 32.58 (C_9), 31.73 (C_4 , C_5), 29.35 (C_3 , C_6), 29.27 and 27.17 (CH_3); MS (ESI): m/z $[\text{M}+\text{H}]^+$ calcd. for $\text{C}_{23}\text{H}_{25}\text{BrO}_3$: 428.10; Found: 429.109.

9-(*m*-tolyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1*H*-xanthene-1,8(2*H*)-dione (3i)

Compound **3i** was synthesized by reacting 3-methylbenzaldehyde (**1i**) (0.118 mL, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from a mixture of ethanol and water to yield white crystals. Yield 320.496 mg, 88%; mp: 206–208 °C; IR (KBr, ν , cm^{-1}): 3060 (aromatic-CH), 2959 and 2870 (aliphatic-CH), 1660 (C=O), 1489, 1363, 1277 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 6.62–7.10 (m, 4H, Ar–H), 4.71 (s, 1H, pyran-CH), 3.74 (s, 3H, Ar- CH_3), 2.13–2.44 (m, 8H, $4\times\text{CH}_2$), 1.08 (s, 6H, $2\times\text{CH}_3$), 0.98 (s, 6H, $2\times\text{CH}_3$), $^{13}\text{C-NMR}$ (100 MHz, CDCl_3) δ_{C} (ppm): 196.70 (C=O), 162.56 ($\text{C}_{4\text{a}}$, $\text{C}_{10\text{a}}$), 159.51, 145.91, 129.10, 121.07, 115.74, 114.51, 112.02, 55.34 (C_2 , C_7), 50.95 (C_4 , C_5), 41.05 (C_9), 32.42 (C_3 , C_6), 32.00 and 29.46 (CH_3), 27.62 (Ar- CH_3); MS (ESI): m/z $[\text{M}+\text{H}]^+$ calcd. for $\text{C}_{24}\text{H}_{28}\text{O}_3$: 364.20; Found: 365.21.

9-([1,1'-biphenyl]-4-yl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1*H*-xanthene-1,8(2*H*)-dione (3j)

Compound **3j** was synthesized by reacting biphenyl-4-carboxaldehyde (**1j**) (0.182 g, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from ethanol to yield white crystals. Yield 375.07 mg, 88%; mp: 211–213 °C; IR (KBr, ν , cm^{-1}): 3061 (aromatic-CH), 2926 and 2949 (aliphatic-CH), 1664 (C=O), 1613, 1489, 1368, 1208 (C=C, C–O); $^{13}\text{C-NMR}$ (100 MHz, CDCl_3) δ_{C} (ppm): 196.90 (C=O), 170.99 ($\text{C}_{4\text{a}}$, $\text{C}_{10\text{a}}$), 169.41, 151.22, 128.20, 127.75, 124.81, 124.52, 118.53, 115.95, 111.23, 50.13 (C_2 , C_7), 43.37 (C_4 , C_5), 41.74 (C_9), 32.54 (C_3 , C_6), 29.40 and 27.44 (CH_3); MS (ESI): m/z $[\text{M}+\text{H}]^+$ calcd. for $\text{C}_{29}\text{H}_{30}\text{O}_3$: 426.22; Found: 427.23.

9-(2,4-dichlorophenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1*H*-xanthene-1,8(2*H*)-dione (3k)

Compound **3k** was synthesized by reacting 2,4-dichlorobenzaldehyde (**1k**) (0.175 g, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from a mixture of ethanol and water to yield white crystals. Yield 367.93 mg, 88%; mp: 253–255 °C; IR (KBr, ν , cm^{-1}): 3059 (aromatic-CH), 2973 and 2884 (aliphatic-CH), 1667 (C=O), 1583,

1480, 1359, 1195 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 7.12–7.35 (m, 3H, Ar–H), 4.94 (s, 1H, pyran-CH), 2.13–2.44 (m, 8H, $4\times\text{CH}_2$), 1.09 (s, 6H, $2\times\text{CH}_3$), 1.00 (s, 6H, $2\times\text{CH}_3$), $^{13}\text{C-NMR}$ (100 MHz, CDCl_3) δ_{C} (ppm): 196.87 (C=O), 163.52 ($\text{C}_{4\text{a}}$, $\text{C}_{10\text{a}}$), 144.75, 138.85, 134.27, 132.96, 130.04, 126.93, 113.50, 50.85 (C_2 , C_7), 40.97 (C_4 , C_5), 32.28 (C_9), 29.52 (C_3 , C_6), 27.55 (CH_3); MS (ESI): m/z $[\text{M} + \text{H}]^+$ calcd. for $\text{C}_{23}\text{H}_{24}\text{Cl}_2\text{O}_3$: 418.11; Found: 419.12.

9-(2-hydroxyphenyl)-3,3,6,6-tetramethyl-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (3l)

Compound **3l** was synthesized by reacting 2-hydroxybenzaldehyde (**1l**) (0.104 mL, 1 mmol) with dimedone (**2a**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from a mixture of ethanol and water to yield white crystals. Yield 333.22 mg, 91%; mp: 205–206 °C; IR (KBr, ν , cm^{-1}): 3443, 3061 (aromatic-CH), 2973 and 2875 (aliphatic-CH), 1632 (C=O), 1569, 1487, 1375 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 7.24–7.59 (m, 4H, Ar–H), 4.56 (s, 1H, pyran-CH), 3.37 (s, 1H, Ar–OH), 2.06–2.54 (m, 8H, $4\times\text{CH}_2$), 1.02 (s, 6H, $2\times\text{CH}_3$), 0.90 (s, 6H, $2\times\text{CH}_3$), $^{13}\text{C-NMR}$ (100 MHz, DMSO) δ_{C} (ppm): 196.80 (C=O), 163.64 ($\text{C}_{4\text{a}}$, $\text{C}_{10\text{a}}$), 144.20, 140.48, 138.72, 129.54, 127.19, 126.87, 114.94, 50.71 (C_2 , C_7), 32.59 (C_4 , C_5), 31.70 (C_9), 29.31 (C_3 , C_6), 27.26 (CH_3); MS (ESI): m/z $[\text{M} + \text{H}]^+$ calcd. for $\text{C}_{23}\text{H}_{26}\text{O}_4$: 366.18; Found: 367.19.

9-(3,4-dimethoxyphenyl)-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (3m)

Compound **3m** was synthesized by reacting 3,4-dimethoxybenzaldehyde (**1m**) (0.166 g, 1 mmol) with 1,3-cyclohexanedione (**2b**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized from methanol to yield white crystals. Yield 318.73 mg, 90%; mp: 215–216 °C; IR (KBr, ν , cm^{-1}): 3061 (aromatic-CH), 2935 and 2823 (aliphatic-CH), 1669 (C=O), 1625, 1517, 1363, 1240 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 6.98 (s, 1H, Ar–CH), 6.66–6.70 (m, 2H, Ar–H), 4.74 (s, 1H, pyran-CH), 3.86 (s, 3H, OCH_3), 3.77 (s, 3H, OCH_3), 1.95–2.67 (m, 12H, $6\times\text{CH}_2$), $^{13}\text{C-NMR}$ (100 MHz, CDCl_3) δ_{C} (ppm): 197.03 (C=O), 164.16 ($\text{C}_{4\text{a}}$, $\text{C}_{10\text{a}}$), 148.54, 147.65, 137.44, 119.78, 117.10, 112.77, 111.00, 56.12 (OCH_3), 55.97 (OCH_3), 37.19 (C_2 , C_7), 31.13 (C_9), 27.36 (C_4 , C_5), 20.53 (C_3 , C_6); MS (ESI): m/z $[\text{M} + \text{Na}]^+$ calcd. for $\text{C}_{21}\text{H}_{22}\text{O}_5$: 354.15; Found: 377.13.

9-(4-bromophenyl)-3,6-diphenyl-3,4,5,6,7,9-hexahydro-1H-xanthene-1,8(2H)-dione (3n)

Compound **3n** was synthesized by reacting 3,4-dimethoxybenzaldehyde (**1n**) (0.185 g, 1 mmol) with 5-phenylcyclohexane-1,3-dione (**2c**) (0.14 g, 2 mmol) following the general procedure described above. The crude product was recrystallized

from methanol to yield white crystals. Yield 482.17 mg, 92%; mp: 202–204 °C; IR (KBr, ν , cm^{-1}): 3063 (aromatic-CH), 2934 and 2889 (aliphatic-CH), 1664 (C=O), 1620, 1480, 1380, 1352 (C=C, C–O); $^1\text{H-NMR}$ (400 MHz, CDCl_3) δ_{H} (ppm): 7.17–7.41 (m, 14H, Ar–H), 3.94 (s, 1H, pyran-CH), 3.29–3.51 (m, 2H, 2 $\times$ CH), 2.59–2.92 (m, 8H, 4 $\times$ CH₂), $^{13}\text{C-NMR}$ (100 MHz, CDCl_3) δ_{C} (ppm): 195.96 and 195.89 (C=O), 163.85 and 163.05 (C_{4a}, C_{10a}), 143.13, 142.13, 142.10, 131.63, 131.45, 131.29, 130.66, 130.60, 130.50, 129.18, 129.09, 127.59, 127.50, 126.98, 126.97, 126.88, 120.72, 116.61, 116.54, 116.29, 44.19 and 43.77 (C₂, C₇), 38.80 and 38.46 (C₄, C₅), 34.97 (C₉), 31.91 and 31.78 (C₃, C₆); MS (ESI): m/z [M+H]⁺ calcd. for C₃₁H₂₅BrO₃: 524.10; Found: 525.10.

CA inhibition studies

Firstly, hCA I and hCA II isozymes were purified using Sepharose®4B-L-tyrosine-*p*-aminobenzenesulfonamide affinity chromatography. Following purification, the isozymes were characterized. Esterase activities were then measured, and the inhibition effects of xanthene derivatives were evaluated by determining IC₅₀ and *K_i* values. All procedures followed protocols established in our previous studies [67–71] and are described in detail in the Supplementary Material.

Drug-likeness and ADMET analysis

Following experimental evaluation, compound **3c**, which exhibited significant inhibitory activity against hCA I, and compound **3n**, which was most potent inhibitor for hCA II, were selected for subsequent pharmacokinetic analysis. Drug-likeness properties were evaluated using Lipinski's Rule of Five, a widely accepted method for predicting the oral bioavailability of drug candidates [72]. Both drug-likeness and ADMET predictions were conducted using the pkCSM web server. This open-access platform offers a reliable interface for assessing pharmacokinetic parameters [73]. To facilitate these predictions, canonical SMILES codes of the selected compounds were generated using OpenBabel, an open-source tool for chemical file format conversion [74].

The drug-likeness and ADMET profiles of AAZ, the reference inhibitor of carbonic anhydrases, had previously evaluated using the same computational methodology [67]. In the current study, the same software tools and parameter settings were applied for compounds **3c** and **3n**. The resulting data were compared with those of AAZ to maintain methodological consistency and establish a reliable reference for comparative analysis.

Molecular docking studies

Molecular docking analyses were performed using AutoDock Vina integrated with the UCSF Chimera 1.18 interface, in accordance with the validated protocol established in our previous studies [75–77]. Grid box parameters were set to include the complete structure of the target proteins. Ligand structures were first modeled in

GaussView [78], and geometry optimizations were carried out in Gaussian 09w using the density functional theory (DFT) method at the B3LYP level with the 3-21G basis set [79].

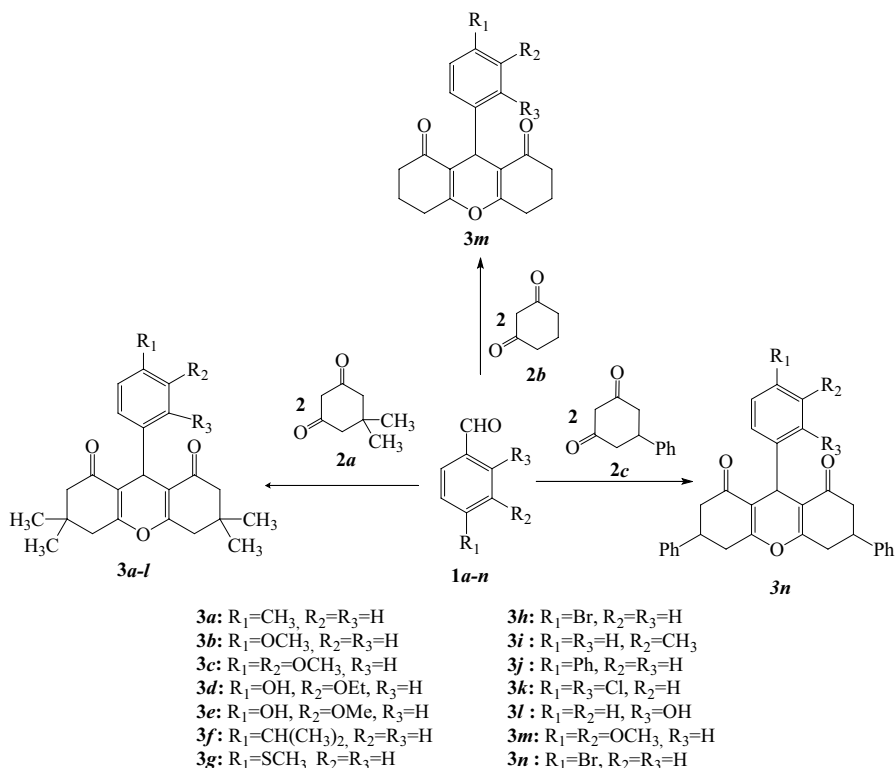
For the docking experiments, crystal structures of human carbonic anhydrase isoenzymes were retrieved from the Protein Data Bank. The 2CAB structure was chosen to represent hCA I, and 1CA2 was selected for hCA II, both due to their high resolutions and low Ramachandran outlier percentages. Protein preparation was conducted by removing water molecules, heteroatoms, co-crystallized ligands, and non-essential chains using BIOVIA Discovery Studio.

Molecular docking analyses employing the standard inhibitor AAZ and the same target proteins were previously reported in detail [67]. In this study, docking simulations for compounds **3c** and **3n** were performed using identical software settings and docking parameters. The resulting ligand–receptor complexes were compared with the validated AAZ docking data to ensure methodological consistency. This strategy facilitated a reliable assessment of the binding conformations and interaction profiles of the new ligands by referencing a well-established inhibitor.

Molecular dynamics simulations

Molecular dynamics (MD) simulations were performed using the GROMACS 2025.3 package with the CHARMM27 force field. Structures of hCA I and hCA II were retrieved from the Protein Data Bank, while ligand topologies for xanthene derivatives were generated via the SwissParam service. Each system was solvated in a triclinic box, maintaining a minimum distance of 1.0 nm between the protein and the box edges, and filled with TIP3P water molecules. Systems were neutralized and adjusted to an ionic strength of 0.1 M NaCl. Energy minimization was performed using the steepest descent algorithm until the maximum force fell below 1000 kJ/mol/nm. Subsequently, a two-step equilibration was implemented: 100 ps in the NVT ensemble at 300 K using the V-rescale thermostat, followed by 100 ps in the NPT ensemble at 1.0 bar using the Parrinello–Rahman barostat.

Production MD simulations were performed for 100 ns with a 2 fs time step. All bonds were constrained using the LINCS algorithm. Short-range electrostatic and van der Waals interactions were calculated with a 1.2 nm cut-off using the Verlet scheme, while long-range electrostatics were determined by the Particle Mesh Ewald (PME) method. Periodic boundary conditions were applied in all directions. Structural stability and residue flexibility of both apo and holo forms were assessed using backbone Root Mean Square Deviation (RMSD) and Root Mean Square Fluctuation (RMSF) analyses, respectively. Trajectories were corrected for periodic boundary conditions and analyzed with GROMACS utilities.



Scheme 1 Synthesis of xanthene derivatives (**3a–n**) from the reaction of cyclic 1,3-diketones (**2a–c**) and aromatic aldehydes (**1a–n**) catalyzed by TFA. Reaction conditions: cyclic 1,3-diketone (2 mmol), aromatic aldehyde (1 mmol), TFA (0.5 mmol), MW (17 atm, 200 W, 100 °C, 10 min)

Results and discussion

Chemistry

In this study, various xanthene derivatives (**3a–n**) were synthesized using a microwave-assisted method, and the results were compared with those obtained by a conventional method. TFA was used as the catalyst during the synthesis studies. Substituted benzaldehyde derivatives (**1a–n**) and cyclic 1,3-diketone derivatives (**2a–c**) were preferred as starting components. Cyclisation reactions were carried out using both conventional and microwave-assisted methods, and the processes were evaluated in terms of reaction time, yield, and reaction progress. The findings revealed that the microwave-assisted method enabled the reactions to proceed more quickly and with a higher yield than the conventional method. Compounds **3j** and **3n** were reported for the first time in the literature.

The reaction scheme (Scheme 1) of the synthesized xanthene derivatives (**3a–n**) and the molecular formulas of each compound (Table 1) are presented below.

Table 1 Synthesized xanthene derivatives (**3a–n**)

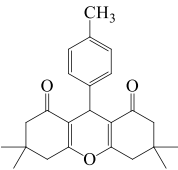
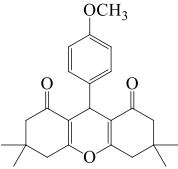
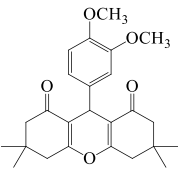
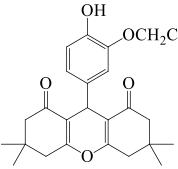
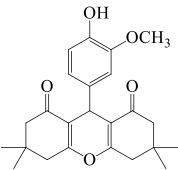
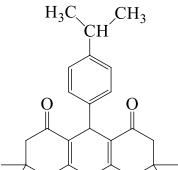
Entry	Product	Time (min)	Yield ^a (%)	mp (°C) Determined	mp (°C) Reported	References
1	 3a	10	93	214–216	213–215	[80]
2	 3b	10	92	246–248	249–250	[81]
3	 3c	10	90	182–184	181–183	[82]
4	 3d	10	92	194–196	194–196	[83]
5	 3e	10	91	224–226	226–228	[84]
6	 3f	10	89	170–172	171–172	[85]

Table 1 (continued)

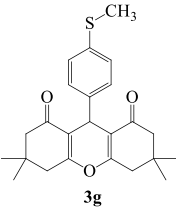
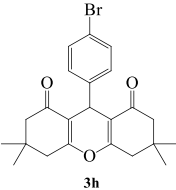
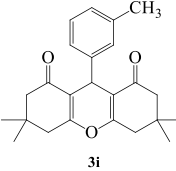
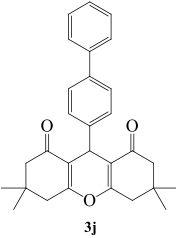
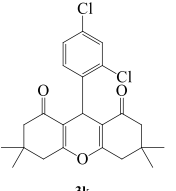
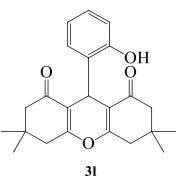
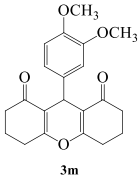
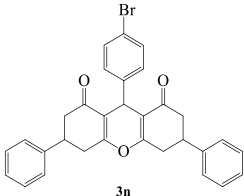
Entry	Product	Time (min)	Yield ^a (%)	mp (°C) Determined	mp (°C) Reported	References
7	 3g	10	91	209–211	209–211	[25]
8	 3h	10	93	240–241	240–242	[86]
9	 3i	10	88	206–208	207–209	[87]
10	 3j	10	89	211–213		
11	 3k	10	88	253–255	253–255	[88]
12	 3l	10	91	205–206	206–207	[33]

Table 1 (continued)

Entry	Product	Time (min)	Yield ^a (%)	mp (°C) Determined	mp (°C) Reported	References
13	 3m	10	90	215–216	216	[89]
14	 3n	10	92	202–204		

The structures of the compounds were confirmed using FT-IR, ¹H-NMR, ¹³C-NMR, and MS techniques. In the IR spectra, aromatic C–H stretching bands appeared in the range 3083–3024 cm^{−1}, while aliphatic C–H stretching vibrations were observed in the range 2973–2819 cm^{−1}. Sharp and strong bands corresponding to the stretching vibrations of the ketone carbonyl group (C=O) were present in the range 1674–1632 cm^{−1}. Additionally, the C=C stretching vibrations of aromatic double bonds were identified as sharp, intense, and multiple bands in the range 1627–1459 cm^{−1}. The stretching vibrations of the –OH group were detected at 3435 cm^{−1} in compound **3d**, 3416 cm^{−1} in compound **3e**, and 3443 cm^{−1} in compound **3l**. The characteristic vibrational bands observed in the IR spectrum are consistent with those of similar xanthene derivatives reported in the literature [90–93], indicating that the synthesis of the compound was successful and the structural confirmation is reliable.

Analysis of the ¹H-NMR spectra of the synthesized compounds (**3a–n**) indicated that aromatic protons produced signals within the δ = 7.59–6.54 ppm range. Singlet signals between δ = 4.94–3.94 ppm were attributed to methine (–CH–) protons in the pyran ring. In the aliphatic region, methylene (–CH₂) protons exhibited multiplet resonance signals in the δ = 2.92–1.95 ppm range. Additionally, methyl groups attached to the xanthene skeleton in the **3a–l** derivatives gave characteristic signals in the δ = 1.09–0.87 ppm range. The observed chemical shift values are consistent with those reported in the literature for comparable xanthene derivatives [94–96]. These findings support the successful synthesis of the compounds and corroborate the proposed molecular structures.

Analysis of the ¹³C-NMR spectra of the synthesized compounds revealed signals in the range δ = 198.26–195.96 ppm, corresponding to the ketone carbonyl (C=O) group. For the **3a–l** derivatives, the double bond carbons (C=C) adjacent to the oxygen atom in the pyran ring exhibited signals within δ = 170.99–162.34 ppm. In the aliphatic region, characteristic signals for methine, methylene, and methyl carbons

were observed in the expected regions (see experimental protocols). On the other hand, in compound **3m**, no signals corresponding to these groups were observed in the aliphatic region, which is attributed to the absence of methyl groups attached to the xanthene skeleton. In compound **3n**, the lack of methyl group signals and the presence of characteristic signals in the aromatic region indicate that phenyl rings have replaced the methyl groups in the aliphatic region. The observed ^{13}C -NMR chemical shift values align with those previously reported for similar xanthene derivatives, thereby providing strong evidence for the structural accuracy of the synthesized compounds [97–99].

Examination of the mass spectra of the synthesized compounds revealed that the experimentally obtained m/z values were in complete agreement with the theoretically calculated values (see experimental protocols). This concordance provides strong evidence for the structural accuracy of the synthesized products, indicating that the molecular masses correspond to the proposed structures. Furthermore, the mass spectral analysis findings align with data from other spectroscopic methods, thereby corroborating the accuracy of the proposed molecular structures.

Inhibition of CA

In this study, major human hCA isoforms (hCA I and hCA II), which play important roles in physiological processes such as pH and electrolyte balance and are therefore associated with disorders such as glaucoma, were selected as the target. The inhibitory effects of synthesized xanthene derivatives on hCA I and hCA II were investigated by purifying both isoforms from human erythrocyte cells using affinity chromatography. hCA I was purified 124.21-fold with a specific activity of 981.33 EU/(mg protein) and a yield of 21.07%, whereas hCA II was purified 233.62-fold with a specific activity of 1845.63 EU/(mg protein) and a yield of 25.47% (Table 2). SDS-PAGE analysis revealed single bands at the expected molecular weights, indicating high purity (Fig. 3).

Examination of the inhibition profiles of the synthesized molecules on hCA I and hCA II isoforms revealed that all compounds generally exhibited moderate inhibition potential. Compared to the control compound AAZ, all compounds except **3n** were weaker inhibitors. The inhibitory potential of **3n** against hCA II was comparable to that of AAZ (see K_i values in Table 3). Sulfonamides are recognized as the most potent inhibitors of hCA isoenzymes. Although other compound classes have been reported in the literature as hCA inhibitors, high inhibitory potency is not always observed in every molecule. This is attributed to the ability of free sulfonamides to coordinate with the Zn^{2+} ion in the active site of hCA [100]. For molecules other than sulfonamides, the strength of inhibitor-enzyme interactions is more strongly influenced by the three-dimensional structure of the molecule. The relatively moderate inhibitory potential of the synthesized molecules can be attributed to the absence of free sulfonamide groups in their structures. In terms of the xanthene scaffold, some compounds contain a 3,3,6,6-tetramethyl group in the xanthene ring structure (**3a–l**), one compound

Table 2 Purification data for hCA I and hCA II isoenzymes

Purification step	Activity (EU/mL)	Total volume (mL)	Protein (mg/mL)	Total protein (mg)	Total activity (EU)	Specific activity (EU/mg protein)	Yield (%)	Purification Factor
Hemolysate	97.82	100.0	12.39	1239	9782	7.90	100.00	1.00
Affinity column (hCA I)	412.5	5.0	0.42	2.10	2060.8	981.33	21.07	124.21
Affinity column (hCA II)	498.32	5.0	0.27	1.35	2491.6	1845.63	25.47	233.62

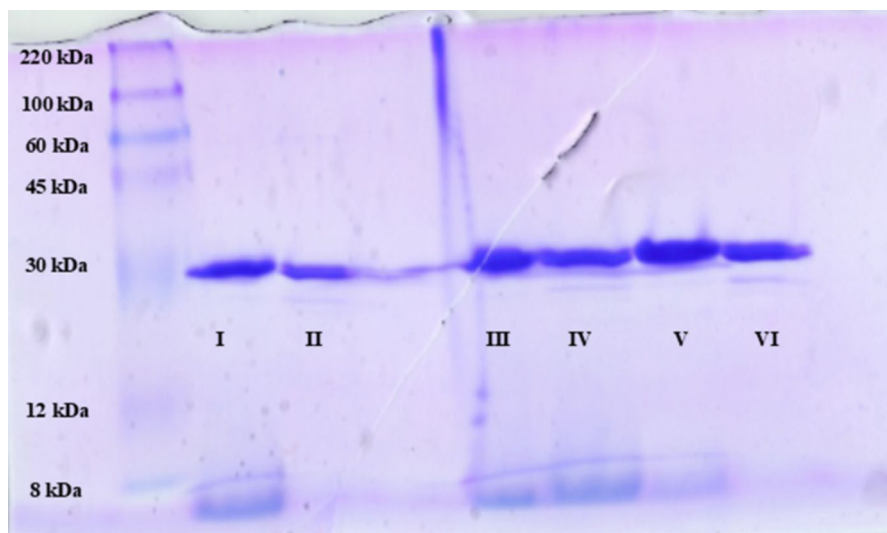


Fig. 3 SDS-PAGE analysis of purified hCA I and hCA II (I–III for hCA I, IV–VI for hCA II)

Table 3 Inhibition data of synthesized xanthene derivatives

Compounds	Esterase IC ₅₀ (μM) ^{a,b}			K _i (μM) ^{a,b}		
	hCA I	hCA II	hCA I	Inhibition type	hCA II	Inhibition type
AAZ^c	0.420	0.310	0.260	Noncompetitive	0.140	Noncompetitive
3a	22.831	12.981	45.800	Competitive	26.039	Noncompetitive
3b	26.287	9.520	8.527	Uncompetitive	10.558	Noncompetitive
3c	7.574	3.279	3.619	Noncompetitive	3.158	Noncompetitive
3d	21.868	19.249	10.541	Noncompetitive	1.825	Uncompetitive
3e	57.549	18.647	32.613	Noncompetitive	1.926	Noncompetitive
3f	NI	NI	–	–	–	–
3g	NI	5.997	–	–	1.880	Uncompetitive
3h	NI	14.485	–	–	3.290	Uncompetitive
3i	29.884	7.692	4.784	Uncompetitive	7.784	Noncompetitive
3j	NI	NI	–	–	–	–
3k	39.003	41.820	–	–	–	–
3l	21.447	13.532	47.789	Noncompetitive	–	–
3m	28.046	6.619	17.313	Noncompetitive	2.080	Noncompetitive
3n	63.018	10.585	12.650	Uncompetitive	0.178	Uncompetitive

NI: No Inhibition

^aMean from three different assays, by a spectrophotometric technique (errors were in the range of $\pm 5\%$ of reported values)

^b $p < 0.0001$ for all analysis

^cAAZ was used as reference compound. Inhibition data was taken from reference [68]

contains a 3,6-diphenyl group (**3n**), and one compound lacks substituents in the xanthene scaffold (**3m**). These structural differences have also influenced the compounds' inhibitory potentials [101].

IC₅₀ values indicate that compound **3n** is approximately six times more selective for hCA II, while compound **3m** is about four times more selective for hCA II. Among the compounds with two methyl substituents at the 3rd and 6th carbon atoms of the xanthene scaffold, compound **3c** demonstrated the highest inhibitory potency for both hCA I and hCA II. Notably, compounds **3g** and **3h** inhibited only the hCA II isoenzyme. Both **3h** and **3n** contain a 4-bromophenyl substituent, which may contribute to this selectivity. According to the IC₅₀ values in Table 3, compounds **3b**, **3c**, **3i**, and **3e** are 3-, 2-, 4-, and 3-times more selective for hCA II, respectively. Compounds **3f** and **3j** did not inhibit both isoforms (Table 3). The isopropylphenyl substituent in compound **3f** and the 1,1'-biphenyl substituent in compound **3j** may hinder inhibitor activity by blocking the hCA–compound interaction through steric effects. The inhibition potentials of compounds **3a**, **3d**, and **3l** for hCA I and hCA II are similar. Although the -Cl substituent was expected to enhance inhibition due to its electronegativity, compound **3k**, which contains a 2,4-dichlorophenyl substituent, was found to be a very weak inhibitor, possibly due to steric hindrance affecting enzyme-inhibitor interaction. The 4-methylphenyl substituted compound **3a** exhibits similar inhibition for both isoforms, whereas the 3-methylphenyl substituted compound **3i** is approximately four times more selective for hCA II, indicating that the position of a small alkyl group can significantly influence inhibition. Compounds containing a methoxy group in the phenyl substituent (**3b**, **3c**, **3e**, and **3m**) display greater selectivity for hCA II (Table 3). Furthermore, compound **3g**, which contains a 4-(methylthio)phenyl substituent and closely resembles the methoxyphenyl structure, is selective only for hCA II, suggesting that substituting the O atom with the bulkier S atom substantially alters isoform selectivity.

Analysis of the equilibrium constant of the inhibition reaction (K_i) indicates that compound **3c** is the most effective inhibitor of hCA I, whereas compound **3n** is the most effective inhibitor of hCA II. Because K_i values were determined using different substrate concentrations, compounds **3c** and **3n**, which exhibited the most effective K_i values, were selected as targets for the in silico component of this study to ensure greater reliability of the results. Compound **3c** inhibits the hCA I isoenzyme via a noncompetitive mechanism, while compound **3n** inhibits the hCA II isoenzyme through an uncompetitive mechanism (Fig. 4). Furthermore, compound **3n** demonstrates an effect comparable to AAZ in terms of effective K_i value. The inhibition potentials of the compounds for hCA I, based on their K_i values, are ranked as follows: **3c** > **3i** > **3b** > **3d** > **3n** > **3m** > **3e** > **3a** > **3l**. For hCA II, the inhibition potentials are ranked as: **3n** > **3d** > **3g** > **3e** > **3m** > **3c** > **3h** > **3i** > **3b** > **3a** (Table 3). The K_i values of the compounds for hCA II are generally more potent than those for hCA I. Some compounds do not have K_i values listed in Table 3. For accurate calculation of the K_i constant and determination of the inhibition mechanism, the inhibitor must reduce enzyme activity by more than 50%. Since compounds **3f**, **3j**, and **3k** do not decrease hCA activity by more than 50%, their K_i constants could not be determined.

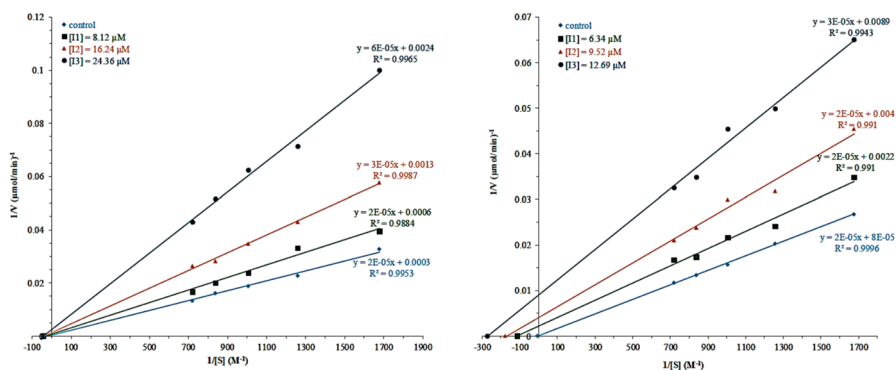


Fig. 4 K_i plots of **3c** for hCA I (left) and **3n** for hCA II (right)

Drug-likeness and ADMET evaluation

The drug-likeness properties of compounds **3c** and **3n** were evaluated according to Lipinski's Rule of Five, as summarized in Table 4. Compound **3c** meets most of the Lipinski criteria, although LogP is at the upper threshold, including molecular weight (410.51 Da) and appropriate numbers of hydrogen bond donors and acceptors. While the LogP value (5.10) is at the upper threshold, it remains within the acceptable range for oral bioavailability.

In contrast, compound **3n** does not meet two of Lipinski's criteria: molecular weight (525.44 Da) and LogP (7.36), both of which exceed the recommended thresholds. These deviations suggest potential challenges for oral absorption and bioavailability. However, the lack of hydrogen bond donors and the presence of an acceptable number of hydrogen bond acceptors indicate a relatively favorable permeability profile.

Compound **3c** exhibits a more favorable profile for oral administration, while compound **3n** requires further evaluation regarding formulation strategies and bioavailability. This assessment aligns with the widely accepted perspective that Lipinski's Rule of Five serves as an empirical guideline rather than an absolute rule, and that compounds outside these parameters may still demonstrate acceptable pharmacokinetic profiles [102, 103].

Table 4 Drug-likeness analysis of **3c**, **3n** and AAZ

Lipinski's five criteria	Accepted range	3c		3n		AAZ	
		Value	Result	Value	Result	Value	Result
Molecular Weight (Da) (MW)	≤ 500	410.51	✓	525.44	✗	194.09	✓
Num. H-bond donors	≤ 5	0	✓	0	✓	1	✓
Num. H-bond acceptors	≤ 10	5	✓	3	✓	6	✓
MLogP	≤ 5	5.1	✓	7.36	✗	2.36	✓

The ADMET profiles of compounds **3c** and **3n** were assessed relative to the reference compound AAZ, as reported in our previous study [67]. Key pharmacokinetic parameters, including absorption, distribution, metabolism, excretion, and toxicity, are presented in Table 5. Based on the absorption data in Table 5, AAZ demonstrates the highest water solubility ($-1.925 \log \text{ mol/L}$) and moderate human intestinal

Table 5 ADMET analysis results obtained by using pkCSM tools

Parameters	Compounds		
Absorption	3c	3n	AAZ
Water solubility ($\log \text{ mol/L}$)	-4.976	-6.307	-1.925
Caco2 permeability ($\log \text{ Papp}$ in 10^{-6} cm/s)	1.569	1.322	-0.305
Human intestinal absorption (HIA+, %)	97.619	97.125	54.854
Skin Permeability ($\log \text{ Kp}$)	-2.938	-2.86	-2.736
P-glycoprotein substrate	No	Yes	No
P-glycoprotein I inhibitor	Yes	Yes	No
P-glycoprotein II inhibitor	Yes	Yes	No
Distribution			
VDss (human) ($\log \text{ L/kg}$)	0.189	-0.089	-0.546
Fraction unbound (human) (Fu)	0	0	0.918
BBB permeability ($\log \text{ BB}$)	-0.723	0.344	-0.868
CNS permeability ($\log \text{ PS}$)	-1.665	-0.518	-4.445
Metabolism			
CYP2D6 substrate	No	No	No
CYP3A4 substrate	Yes	Yes	No
CYP1A2 inhibitor	No	No	No
CYP2C19 inhibitor	Yes	No	No
CYP2C9 inhibitor	No	No	No
CYP2D6 inhibitor	No	No	No
CYP3A4 inhibitor	Yes	No	No
Excretion			
Total Clearance ($\log \text{ mL/min/kg}$)	0.258	-0.213	0.364
Renal OCT2 substrate	No	No	No
Toxicity			
AMES toxicity	No	No	No
Max. tolerated dose (human) ($\log \text{ mg/kg/day}$)	0.303	0.77	1.27
hERG I inhibitor	No	No	No
hERG II inhibitor	No	Yes	No
Oral Rat Acute Toxicity (LD50) (mol/kg)	2.874	2.472	1.861
Oral Rat Chronic Toxicity (LOAEL) ($\log \text{ mg/kg}_{\text{bw/day}}$)	2.103	2.128	2.605
Hepatotoxicity	No	No	No
Skin Sensitization	No	No	No
T. Pyriformis toxicity ($\log \text{ ug/L}$)	0.374	0.355	0.136
Minnow toxicity ($\log \text{ mM}$)	0.615	-2.362	3.915

absorption (54.85% HIA). In contrast, **3c** (water solubility $-4.976 \log \text{ mol/L}$, HIA 97.62%) and **3n** (water solubility $-6.307 \log \text{ mol/L}$, HIA 97.13%) display significantly higher absorption rates [104]. Nevertheless, **3n**'s low solubility may limit its bioavailability despite its high absorption. This poor aqueous solubility, likely associated with increased lipophilicity, may hinder its formulation and reduce effective systemic exposure under physiological conditions [105].

Regarding distribution, compounds **3c** and **3n** exhibit moderately poor profiles, with volume of distribution at steady state (VD_{ss}) values of 0.189 and $-0.089 \log \text{ L/kg}$, respectively. In contrast, AAZ demonstrates limited distribution, with a VD_{ss} of $-0.546 \log \text{ L/kg}$. Compound **3n** shows positive blood–brain barrier (BBB) permeability (0.344 log BB), indicating potential for central nervous system activity. Conversely, **3c** ($-0.723 \log \text{ BB}$) and AAZ ($-0.868 \log \text{ BB}$) display poor BBB penetration, which likely restricts their central nervous system effects [73]. Although the BBB permeability of **3n** offers potential benefits for central nervous system-targeted applications, it may also elevate the risk of off-target central effects. This consideration should inform subsequent optimization studies.

AAZ does not interact with CYP3A4, thereby reducing the risk of drug–drug interactions. In contrast, both **3c** and **3n** are CYP3A4 substrates, which may increase the likelihood of metabolic interactions. Furthermore, **3c** acts as a CYP3A4 inhibitor, which could complicate polypharmacy. Excretion data demonstrate that AAZ has the highest total clearance ($0.364 \log \text{ mL/min/kg}$), indicating rapid elimination. In comparison, **3c** exhibits moderate clearance ($0.258 \log \text{ mL/min/kg}$), whereas **3n** shows slow elimination ($-0.213 \log \text{ mL/min/kg}$), potentially leading to accumulation in the body [106]. This slow clearance profile of **3n**, when combined with its lipophilic nature, may further increase the risk of bioaccumulation and prolonged systemic exposure.

Finally, toxicity assessments indicate that none of the compounds exhibit hepatotoxicity, as demonstrated in Table 5. However, compound **3n** functions as a hERG II inhibitor, which may increase the risk of cardiac-related side effects. This finding represents a notable safety concern, as hERG inhibition is commonly associated with potential cardiotoxicity and arrhythmogenic risk in drug candidates [107]. In contrast, AAZ demonstrates lower acute toxicity (LD₅₀: 1.861 mol/kg) and a more favorable overall toxicity profile. While compound **3n** demonstrates promising biological activity, its pharmacokinetic and toxicity limitations suggest that additional structural optimization is necessary to enhance its drug-likeness profile. Conversely, compound **3c** exhibits a more balanced ADMET profile, supporting its identification as a more suitable lead candidate.

Molecular docking studies

In silico methods for optimizing drug candidate molecules offer substantial advantages in terms of time and cost efficiency. Among these approaches, molecular docking is commonly utilized to predict the binding mechanisms of candidate compounds with target biological systems and to analyze these interactions at the molecular level [108–110]. Previous studies indicate that both the number and types of

interactions between the ligand and receptor are critical in determining the stability of the resulting complex [111].

In this study compared the binding interactions of ligands **3c** and **3n**, complexed with receptors 2CAB and 1CA2, to those of the standard inhibitor AAZ as reported previously [67]. AAZ's hydrogen bonding and hydrophobic interactions with receptor binding sites served as a reference for assessing the binding behaviors of **3c** and **3n**. Both candidate compounds demonstrated substantial hydrophobic interactions within the binding pockets of the hCA receptors. These hydrophobic interactions likely contribute to the stabilization of the **3c**@2CAB and **3n**@1CA2 complexes, thereby increasing binding affinity and inhibitory potential [112]. Moreover, the binding sites of both compounds overlapped with the primary binding pocket of the hCA receptors, which corresponds to the site where the reference inhibitor AAZ binds. These pockets contain critical amino acid residues, specifically His94, His96, and His119, located within the active site of both 2CAB and 1CA2 receptors. These residues are recognized as key contributors to inhibitor binding. Importantly, His94 is one of the histidine residues that coordinate the catalytic Zn^{2+} ion in the active site of carbonic anhydrase. Although the synthesized xanthene derivatives do not contain a classical zinc-binding group (ZBG), their interaction with His94 may indirectly affect the Zn^{2+} coordination environment. The detailed intermolecular interactions of the optimal docking poses for the **3c**@2CAB and **3n**@1CA2 complexes, as well as their calculated binding affinities, are summarized in Table 6.

Three-dimensional (3D) and two-dimensional (2D) interaction diagrams depicting these molecular contacts are shown in Figs. 5 and 6, respectively. As illustrated in Fig. 5, the **3c** ligand is stabilized within the active site of hCA I through a combination of hydrogen bonding, hydrophobic interactions, and van der Waals contacts.

Similarly, Fig. 6 demonstrates that the **3n** ligand adopts a stable binding conformation within the hCA II active site, supported by extensive hydrophobic interactions and key contacts with residues such as His94.

These structural observations further confirm that both ligands occupy the canonical active site cavity of carbonic anhydrase, overlapping with the binding region of the reference inhibitor AAZ. In contrast to the reference inhibitor AAZ, which directly coordinates with the Zn^{2+} ion (2.39–3.02 Å), compounds **3c** and **3n** do not exhibit direct ligand– Zn^{2+} interactions. Rather, both compounds interact with His94, a residue directly coordinated to Zn^{2+} (His94– Zn^{2+} distances of 0.8 Å and 2.13 Å for 2CAB and 1CA2, respectively), suggesting a second-shell or indirect inhibition mechanism. These results indicate that the synthesized compounds do not function as classical Zn^{2+} -binding inhibitors. Instead, they inhibit by affecting the His94– Zn^{2+} coordination network. This mechanism aligns with non-classical carbonic anhydrase inhibition, in which ligand binding within the active site indirectly influences the catalytic metal center.

Binding affinity values are critical indicators for evaluating the effectiveness of candidate compounds in receptor-ligand complexes. As presented in Table 6, the binding affinities for the complexes of **3c** with 2CAB and **3n** with 1CA2 were calculated as –7.4 kcal/mol and –10.5 kcal/mol, respectively. In contrast, the binding affinities of AAZ with 2CAB and 1CA2 receptors were determined to be –6.0 kcal/mol and –6.1 kcal/mol, respectively [67]. These results suggest that

Table 6 Summary of molecular docking results for the **3e@2CAB** and **3n@1CA2** complexes

Complex	ΔG (kcal/mol)	Hydrogen Bond interactions (Å)	Hydrophobic interaction (Å)	Other interaction	Zn ²⁺ interaction
3e@2CAB	− 7.4	Carbon H-Bond: His67(2.37)	Pi-Pi T-shaped: His67(5.67)	van der Waals: Ile60, Val62, Asn69, His92, Leu141, His94–Zn(0.8)	Indirect (via His94, no direct Zn ²⁺ coordination)
			Pi-Alkyl: Phe91(5.42), His94(5.16), His200(4.28), Tyr204(4.95)		
			Alkyl: Ala132(4.26), Leu131(4.73), Ala135(4.04–4.48–3.53), Leu198(4.1–4.61–3.39), Pro202(4.45)		
AAZ@2CAB*	− 6.0	Conventional H-Bond: Ser193(2.70), Thr195(2.27)	Pi-Pi T-shaped: His90(5.16)	Metal Acceptor: Zn(3.02) Pi-sulfur: His90(5.06–5.14)	Direct (Zn ²⁺ coordination)
		Carbon H-Bond: His90(2.76)	Alkyl: Leu194(4.10)		
		Conventional H-Bond: Thr200(2.76)			
3n@1CA2	− 10.5		Pi-Sigma: Ile91(3.94)	van der Waals: Glu69, Gln92, His119, Thr199, Pro201, Trp209 His94–Zn(2.13)	Indirect (via His94, no direct Zn ²⁺ coordination)
			Pi-Pi T-shaped: His94(5.01)		
			Alkyl: Val121(4.99), Leu198(5.46), Ile91(5.42), Val135(4.45), Pro202(3.24), Leu204(3.75)		
AAZ@1CA2*	− 6.1		Pi-Alkyl: Phe131(5.5), Val135(5.42), Pro202(3.66), Val121(5.04), Val143(5.44), Leu198(5.44)	Metal Acceptor: Zn(2.39) Pi-sulfur: His96(5.72)	Direct (Zn ²⁺ coordination)
		Conventional H-Bond: Asn67(2.52), Thr199(3.11–2.73), Thr200(2.70–3.06)	Pi-Pi T-stacked: His94(4.02)		
		Pi-Donor H-Bond: Thr200(3.76)			

*From reference[67]

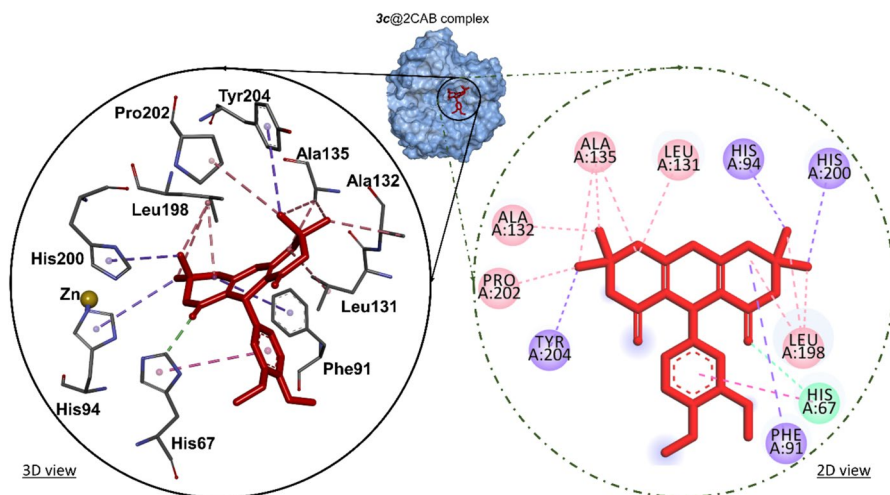


Fig. 5 The 3D and 2D interaction diagrams of the best binding pose of **3c** within the active site of hCA I are presented, respectively

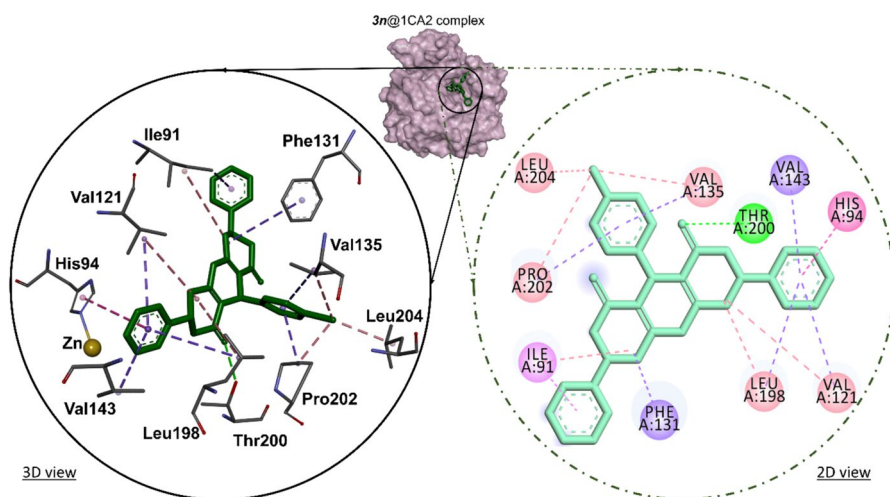


Fig. 6 The 3D and 2D interaction diagrams of the best binding pose of **3n** within the active site of hCA II are presented, respectively

the receptor-ligand complexes formed by **3c** and **3n** possess greater stability than those formed with AAZ. The results indicate that the investigated compounds function as non-classical carbonic anhydrase inhibitors, exhibiting relatively strong binding affinity and a potentially indirect mechanism of action mediated by key active-site residues, such as His94.

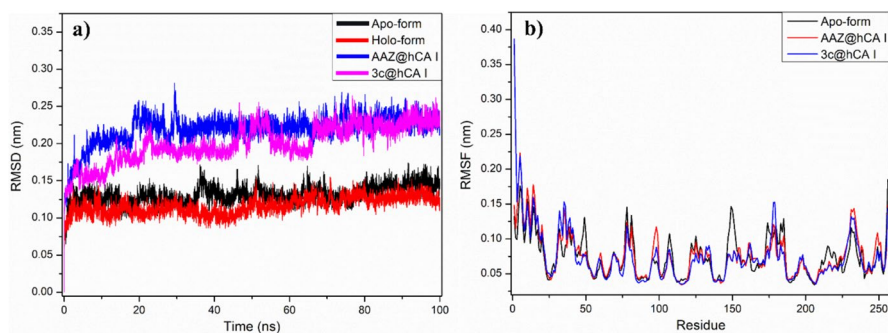


Fig. 7 RMSD (a) and RMSF (b) profiles of hCA I in apo form, holo form, AAZ-bound, and **3c** ligand-bound systems obtained from 100 ns molecular dynamics simulations

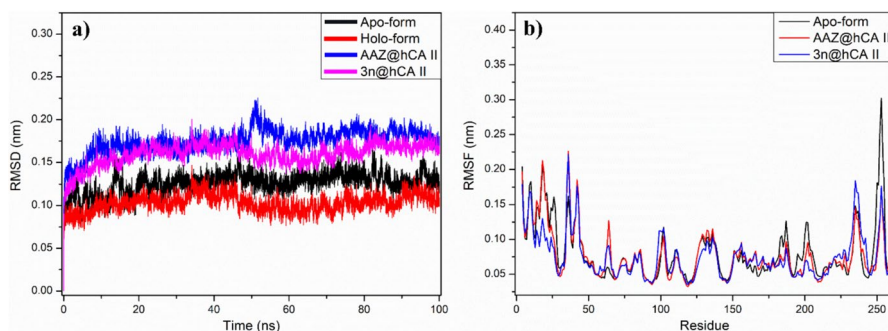


Fig. 8 RMSD (a) and RMSF (b) profiles of hCA II in apo form, holo form, AAZ-bound, and **3n** ligand-bound systems obtained from 100 ns molecular dynamics simulations

Molecular dynamics simulation

Molecular dynamics (MD) simulations were conducted for 100 ns to assess the structural stability and dynamic behavior of free (apo) proteins, holo forms, and ligand-bound complexes, including the standard inhibitor AAZ and the synthesized compounds **3c** and **3n**. Analyses were performed separately for hCA I (2CAB) and hCA II (1CA2), with the corresponding results shown in Figs. 7 and 8, respectively. RMSD and RMSF analyses were used to evaluate global conformational stability and residue-level flexibility.

For the hCA I protein (Fig. 7), the RMSD profiles demonstrate that all systems reached equilibrium following an initial equilibration phase. The apo form exhibited relatively stable behavior with moderate fluctuations between 0.13 and 0.16 nm. The holo form showed slightly lower RMSD values, suggesting that ligand binding contributes to structural stabilization. The AAZ@hCA I complex exhibited RMSD values similar to those of the holo form, reinforcing its function as a reference inhibitor. The **3c**@hCA I complex exhibited somewhat higher RMSD values (approximately

0.20–0.25 nm), which may indicate minor conformational adjustments upon ligand binding. Nevertheless, these fluctuations remained within acceptable limits, confirming that the protein's overall structural integrity was maintained.

The RMSF analysis of hCA I revealed similar patterns of residue flexibility across all systems. Elevated fluctuations were primarily detected in loop regions and terminal residues, which are generally more flexible. The analysis focused on the active site residues His94, His96, and His119. These residues did not show significant RMSF peaks and exhibited lower fluctuations than the highly flexible regions. This finding indicates that the active site remains stable throughout the simulation and that ligand binding does not substantially alter the dynamic behavior of these critical residues.

The RMSD profiles for the hCA II protein (Fig. 8) show behavior similar to that of hCA I, with all systems reaching equilibrium after an initial equilibration phase. The apo form maintained stable RMSD values with moderate fluctuations, whereas the holo form exhibited a slight increase in stability. The AAZ@hCA II complex presented RMSD values comparable to the holo form, reinforcing its role as a stabilizing reference inhibitor. In contrast, the **3n**@hCA II complex showed slightly higher RMSD values compared to the apo and AAZ-bound systems, suggesting minor conformational changes upon ligand binding. Nevertheless, all systems sustained stable trajectories throughout the simulation period.

The RMSF analysis for hCA II demonstrated consistent residue flexibility profiles across all systems. Similar to hCA I, increased fluctuations were predominantly observed in loop regions, whereas active-site residues remained stable. Specifically, His94, His96, and His119 did not exhibit significant fluctuation peaks and maintained low RMSF values, indicating that the structural integrity of the catalytic region is preserved in the presence of both the standard inhibitor and the investigated compounds.

Comparative evaluation of apo, holo, and ligand-bound systems demonstrates that ligand binding does not destabilize the proteins. The standard inhibitor AAZ exhibits stabilizing behavior similar to the holo form, whereas compounds **3c** and **3n** show slightly increased but acceptable fluctuations, likely reflecting minor conformational adaptations. Consistency between RMSD and RMSF analyses indicates that these ligands maintain stable interactions with both hCA I and hCA II, without compromising the integrity of key functional regions. These findings align with docking results and suggest a potential for stable binding. However, additional experimental studies are necessary to confirm inhibitory activity and to elucidate their therapeutic relevance.

Conclusion

Studies on the inhibition of carbonic anhydrases are critical for understanding enzyme inhibition mechanisms and for developing novel pharmacological agents. In this research, a series of substituted xanthene derivatives was synthesized rapidly using microwave-assisted methods, and their structures were characterized. These derivatives exhibited moderate inhibitory activity against hCA I and hCA II

isoenzymes. The influence of modifications to the xanthene ring and the nature of substituent groups on inhibitory activity was systematically examined. Notably, **3c** and **3n** demonstrated potent inhibitory effects against hCA I and hCA II, respectively. Enzyme-inhibitor interactions were analyzed at the molecular level through *in silico* approaches. ADMET analysis revealed a favorable pharmacokinetic profile for **3c**, whereas **3n** requires further evaluation for certain parameters. Molecular docking and 100 ns molecular dynamics simulations confirmed robust, stable interactions between both ligands and the key residues His94, His96, and His119, thereby maintaining the structural integrity of the complexes. The combined docking and molecular dynamics results indicate that both compounds interact with the active-site cavity without direct coordination to the catalytic Zn²⁺ ion, suggesting a non-classical inhibition mechanism mediated by key residues such as His94. This represents a significant deviation from conventional zinc-binding carbonic anhydrase inhibitors and highlights a novel interaction mode within this enzyme class. The findings indicate that compounds **3c** and **3n** are selective and stable hCA inhibitors and are promising candidates for further experimental validation. The integration of enzymatic, computational, and pharmacokinetic analyses in this study provides a comprehensive framework for the rational design of next-generation carbonic anhydrase inhibitors. The observed non-classical inhibition mode may offer improved selectivity and reduced off-target effects compared to classical zinc-binding inhibitors. However, further structural optimization, particularly for compound **3n**, is necessary to enhance its drug-likeness and safety profile.

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Data availability Data will be made available on request.

Declarations

Conflict of interest The authors state that there are no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Authors and Affiliations

Gönül Bükülmez¹ · Şüheda Özkul² · Ekrem Tunca² · Alpaslan Bayrakdar³ · Rahmi Kasımoğulları¹

✉ Ekrem Tunca
ekrem.tunca@dpu.edu.tr

✉ Rahmi Kasımoğulları
rahmi.kasimogullari@dpu.edu.tr

Gönül Bükülmez
gonul.bukulmez@dpu.edu.tr

Şüheda Özkul
suheda.ozkul@dpu.edu.tr

Alpaslan Bayrakdar
alpaslan.bayrakdar@igdir.edu.tr

¹ Faculty of Arts and Sciences, Department of Chemistry, Dumlupınar University, 43100 Kütahya, Turkey

² Faculty of Arts and Sciences, Department of Biochemistry, Dumlupınar University, 43100 Kütahya, Turkey

³ Vocational School of Higher Education for Healthcare Services, Iğdır University, 76100 Iğdır, Turkey